2023 대학생 창작 모빌리티 경진대회 - 무인모빌리티 부문

Stier | 연구계획서

HONGIK UNIV.

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01. INTRODUCTION

About Team / Research Environment

01. INTRODUCTION About Team / Research Environment

Stier

2023 대학생
창작 모빌리티 경진대회
홍익대학교

TEAM LEADER 김영휘 기계·시스템디자인공학과

CONTROL & GPS

팀장 최우진 기계·시스템디자인공학과

팀원 임효근 캠퍼스자율전공학부

팀원 신유재 컴퓨터공학과

LiDAR

팀장 서민경 기계·시스템디자인공학과

팀원 김도현 기계·시스템디자인공학과

VISION

팀장 오준혁 기계·시스템디자인공학과

팀원 이재준 컴퓨터공학과

01. INTRODUCTION

About Team / **Research Environment**

Development Environment

LabVIEW

ubuntu

ROS

python

C++

Test Environment

FMTC Future Mobility Technology Center

서울대학교 시흥캠퍼스
미래모빌리티기술센터

02. RESEARCH GOAL

Keyword / Problem Analysis / Improvements

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Keyword / Problem Analysis / Improvements

“ Safety & Stability ”

02. RESEARCH GOAL

Keyword / **Problem Analysis** / Improvements

Problem 1

불안정한 GNSS 신호로 인한
차선 이탈

Problem 2

객체 구별 과정의 부재로 인한
트랙 이탈

Problem 3

진동 및 햇빛으로 인한
카메라 노이즈 발생

02. RESEARCH GOAL

Keyword / Problem Analysis / **Improvements**

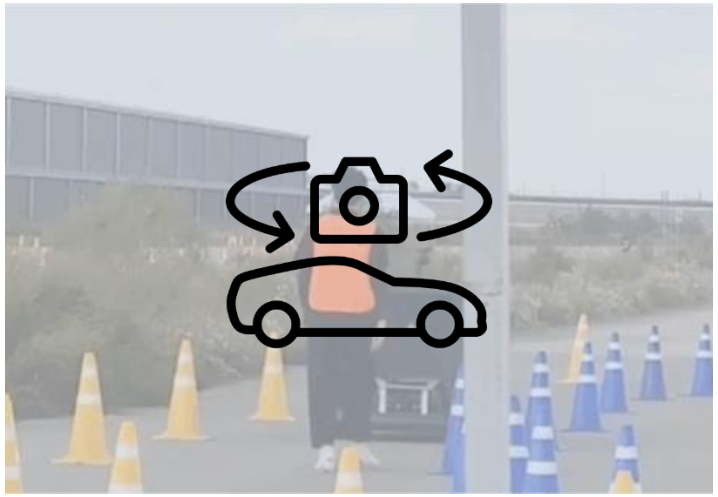
Safety & Stability

Problem 1

불안정한 GNSS 신호로 인한
차선 이탈

IMU
Extended Kalman Filter
Lane Keeping Assist System

Problem 2



객체 구별 과정의 부재로 인한
트랙 이탈

LiDAR & VISION
Sensor Fusion

Problem 3

진동 및 햇빛으로 인한
카메라 노이즈 발생

Machine Vision Camera
(MVC)

03. RESEARCH PLAN

Architecture / Perception / Decision / Control

03. RESEARCH PLAN

Architecture / Perception / Decision / Control

- *H/W Architecture*

Computer

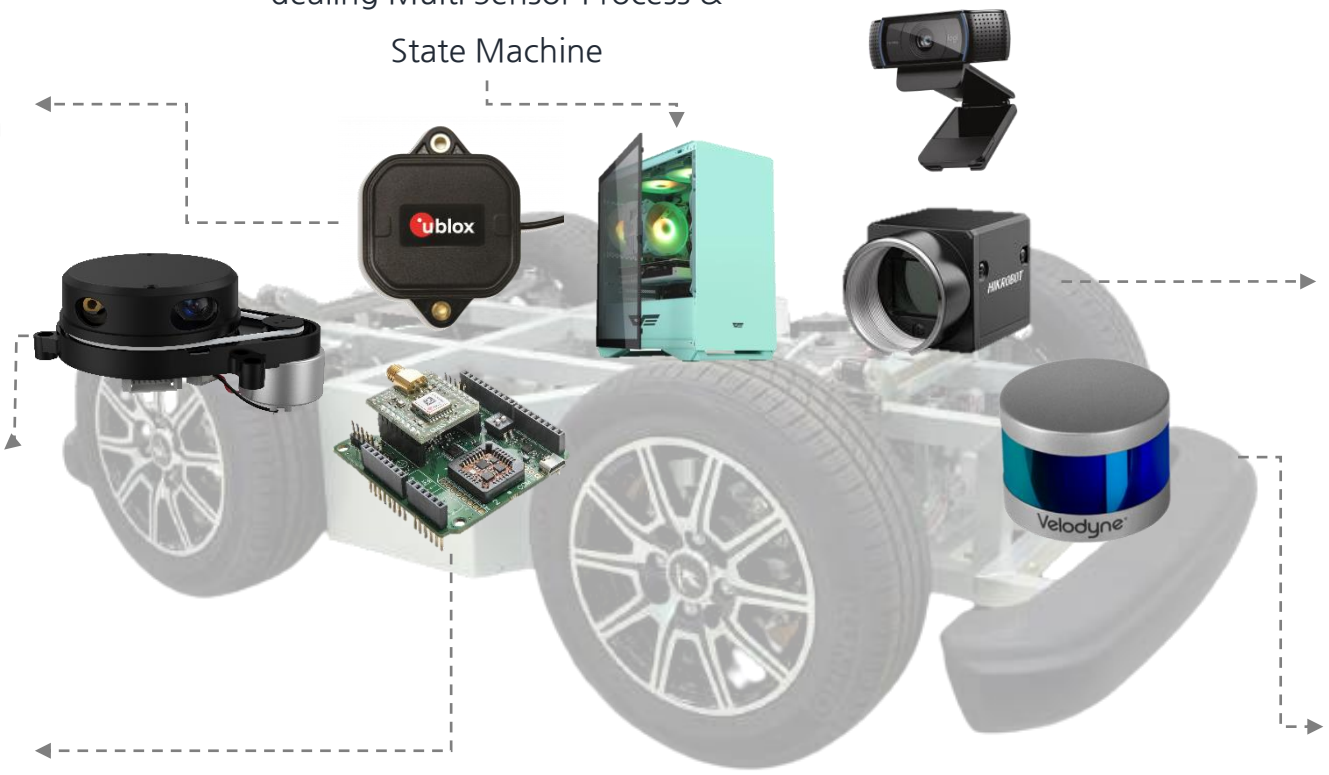
Central Control Computer for dealing Multi Sensor Process & State Machine

Logitech C920

Webcam for Object Detection

ZED-F9P

GNSS reception for Navigation Driving



MVC

Machine Vision Camera for Lane Detection

YDLiDAR X4 Pro

2D LiDAR for Rear Obstacle Detection

MTi-7-0I-DK

Track an object's motion and posture

Velodyne VLP-16

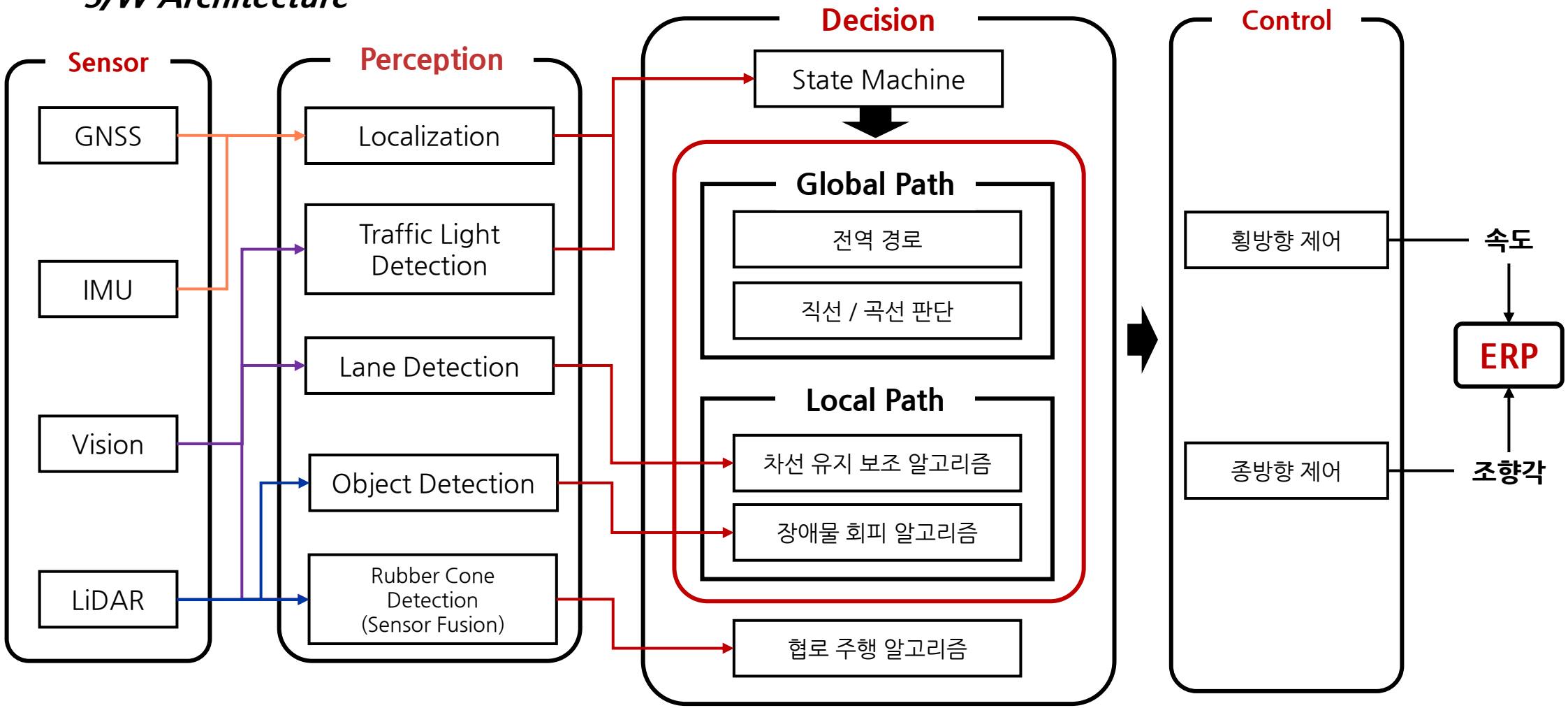
3D LiDAR for Forward Obstacle Detection

ERP-42

03. RESEARCH PLAN

Architecture / Perception / Decision / Control

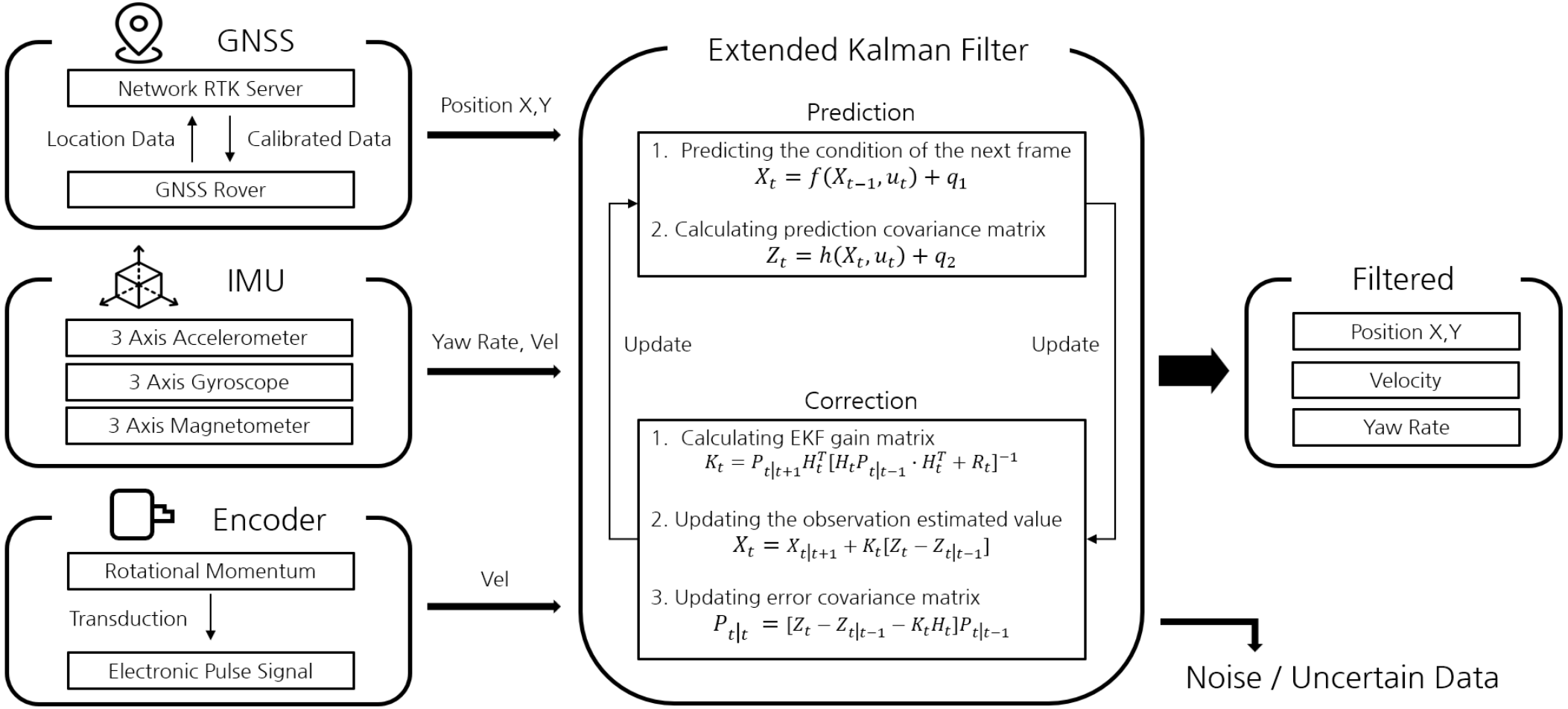
- *S/W Architecture*



03. RESEARCH PLAN

Architecture / **Perception** / Decision / Control

- *Localization*



03. RESEARCH PLAN

Architecture / **Perception** / Decision / Control

- Global Path

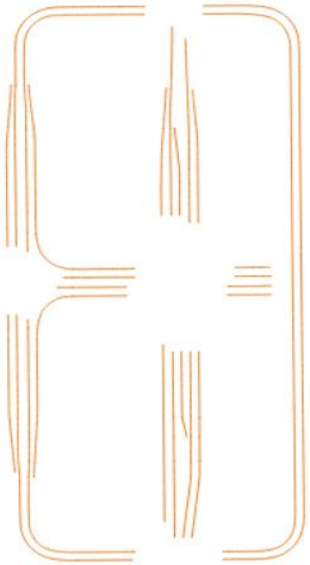
High Definition Map

Filtered Position X,Y

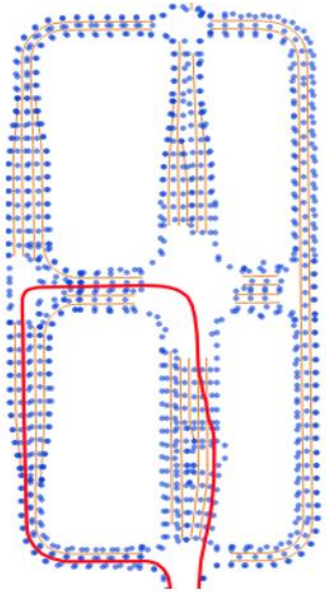
WGS84 ⇒ UTM 52N

K-City

차선 Node



경로 Link



Global path

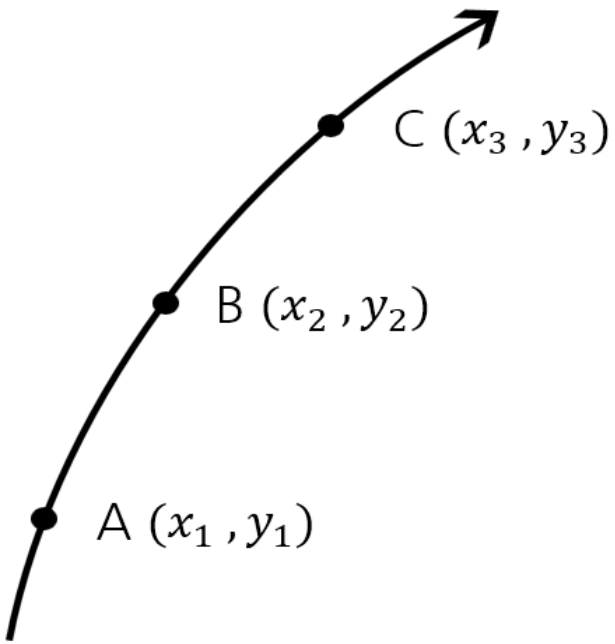
316799.030230	4158127.052456
316799.021641	4158127.063747
316799.004217	4158127.075234
316799.003972	4158127.064137
316798.995138	4158127.064332
316798.994893	4158127.053236
316798.994158	4158127.019947
316799.002503	4158126.997560
316799.011337	4158126.997365
316798.993178	4158126.975562
316799.002013	4158126.975367
316798.957106	4158126.943054
316798.939437	4158126.943444
316798.942867	4158126.943527
316798.943182	4158126.946201

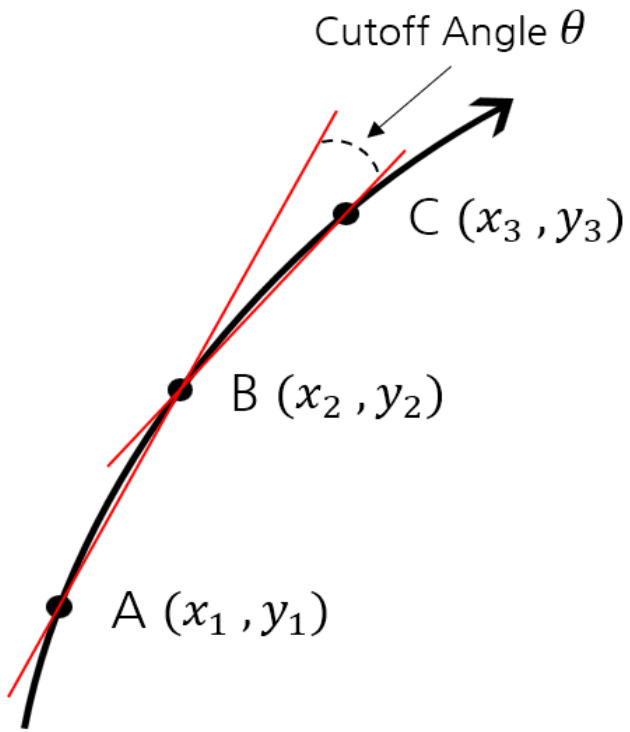
RDDF

03. RESEARCH PLAN

Architecture / **Perception** / Decision / Control

- Straight / Curve Classification





$$\overrightarrow{AB} = (x_2 - x_1) + (y_2 - y_1)$$

$$\overrightarrow{BC} = (x_3 - x_2) + (y_3 - y_2)$$

$$\theta = cos^{-1}\left(\frac{\overrightarrow{AB} \cdot \overrightarrow{BC}}{|\overrightarrow{AB}||\overrightarrow{BC}|}\right)$$

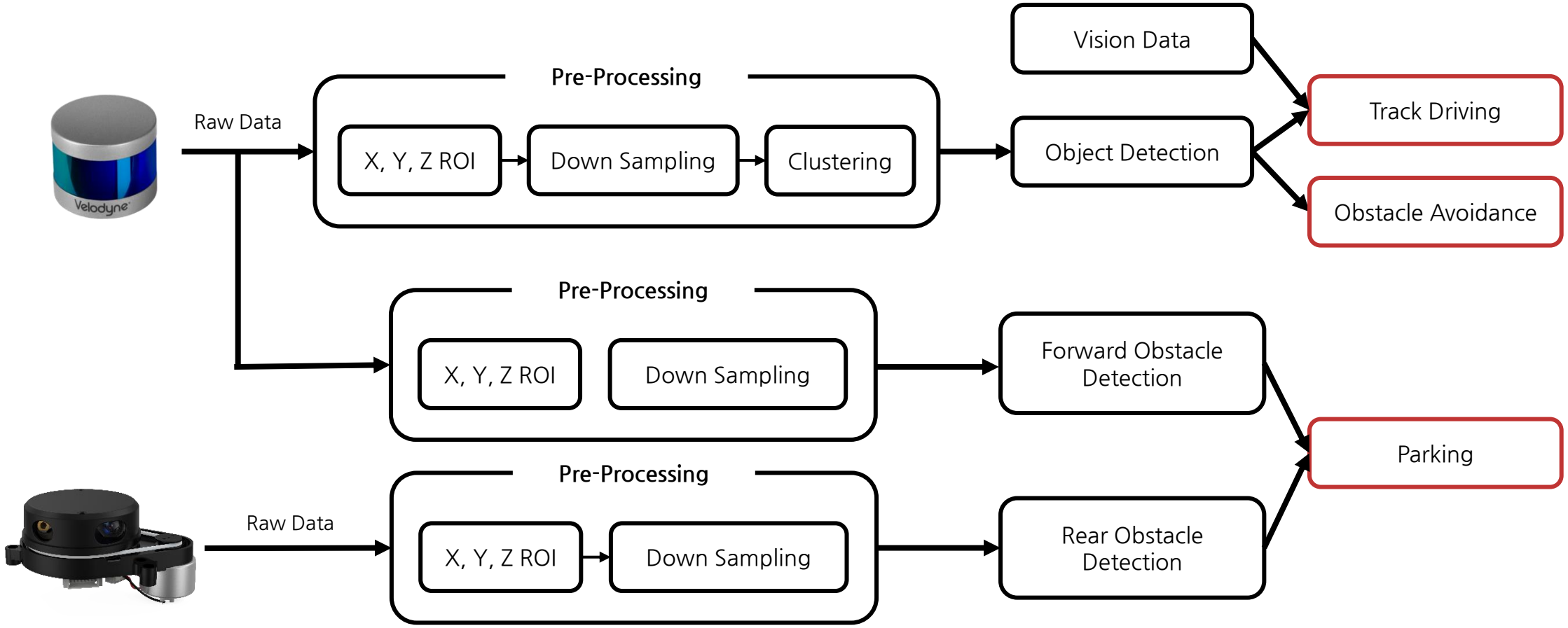
Cutoff Angle θ

- $< 1.5°$ 직선
- $\geq 1.5°$ 곡선1
- $\geq 4.5°$ 곡선2
- $\geq 7.5°$ 곡선3
- ⋮

03. RESEARCH PLAN

Architecture / **Perception** / Decision / Control

- Object Detection



03. RESEARCH PLAN

Architecture / **Perception** / Decision / Control

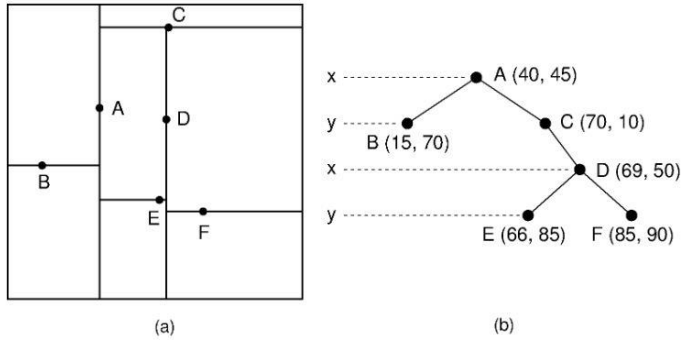
- Pre-Processing for Object Detection

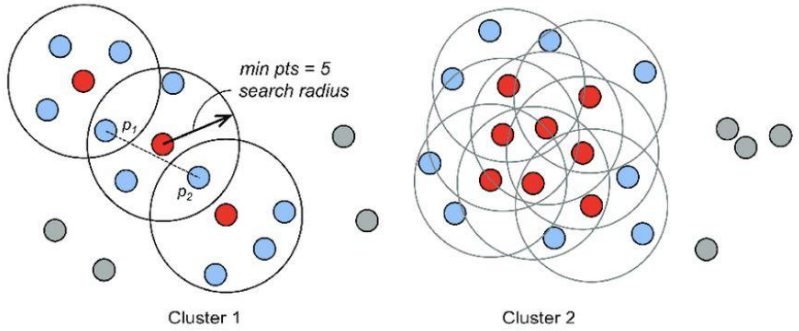
1. Raw Data → **2. X, Y, Z축 ROI 설정** → **3. Down Sampling**

From LiDAR

Voxel Grid

4. Clustering





KD-Tree를 활용한 2D 공간 분할

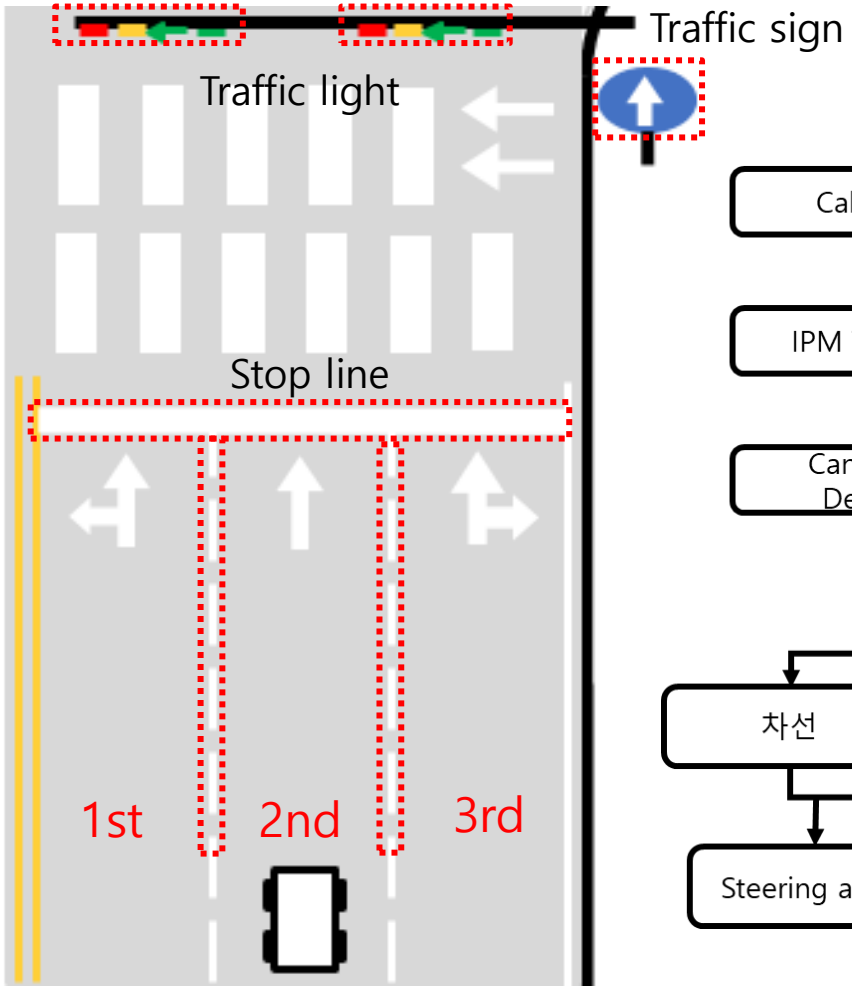
밀도 기반 **DBSCAN**

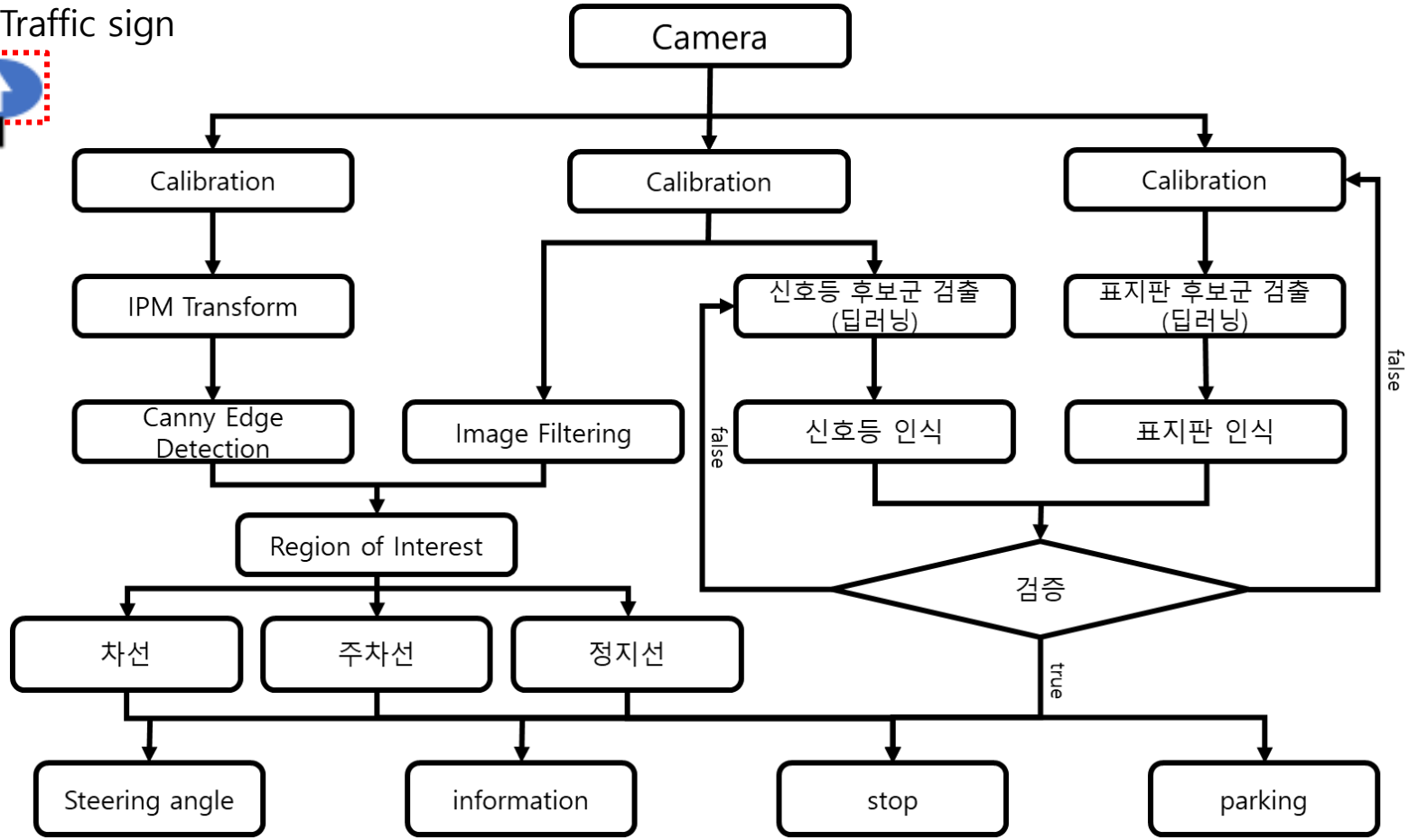
➔ **연산량** 감소

03. RESEARCH PLAN

Architecture / **Perception** / Decision / Control

- *Vision Architecture*





03. RESEARCH PLAN

Architecture / **Perception** / Decision / Control

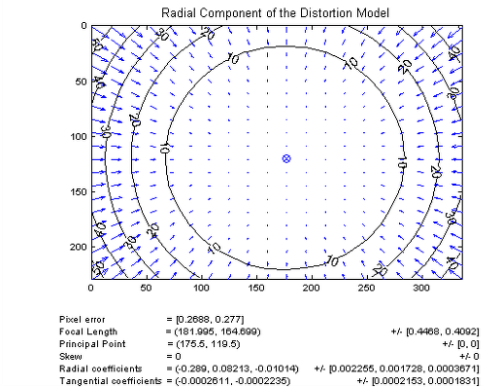
- Improve

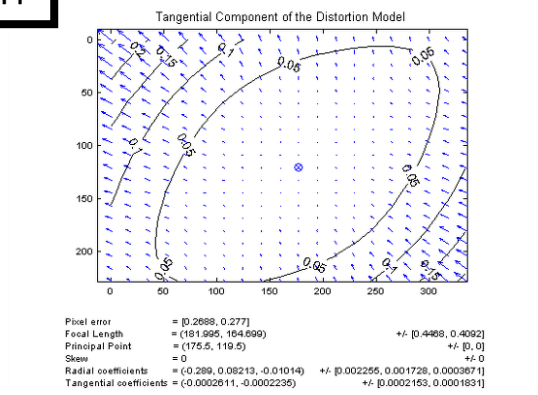
MVC Machine Vision Camera		High FPS	Wide FOV	Global shutter	RGB → Low light reception
Webcam HD pro Logitech C920		Low FPS	Narrow FOV	Rolling shutter	Gray -> High light reception

03. RESEARCH PLAN

- *Calibration*

Inner param





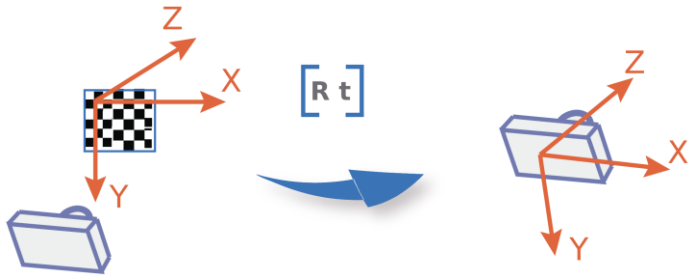
방사형 왜곡

$$x_{rd} = x_n(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$
$$y_{rd} = y_n(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$

접선 왜곡

$$x_{td} = x_{rd} + [2p_1 y_n + p_2(r^2 + 2x_n^2)]$$
$$y_{td} = y_{rd} + [p_1(r^2 + 2y_n^2) + 2p_2 x_n]$$

Outer param



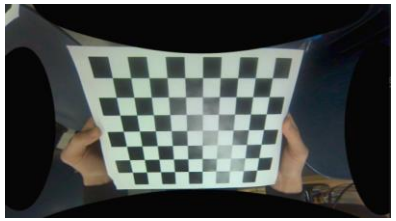
접선 왜곡

$$P_c = R(P_o - T)$$

$$\begin{bmatrix} x_{p_d} \\ y_{p_d} \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & \text{skew_c}f_x & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{n_d} \\ y_{n_d} \\ 1 \end{bmatrix}$$

역산 식

(Calibration 전)



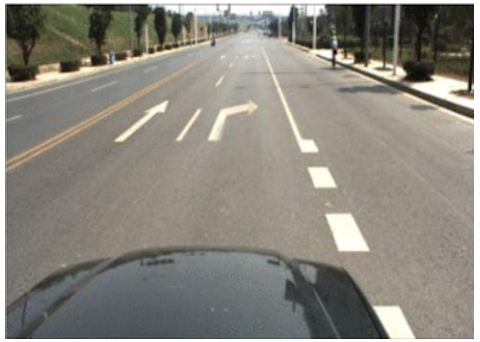
(Calibration 후)

03. RESEARCH PLAN

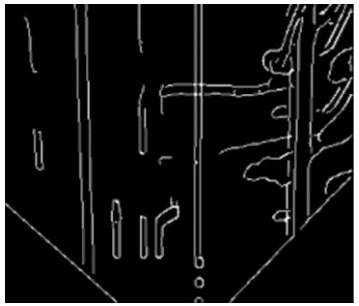
Architecture / **Perception** / Decision / Control

- *Image Processing*

Original image

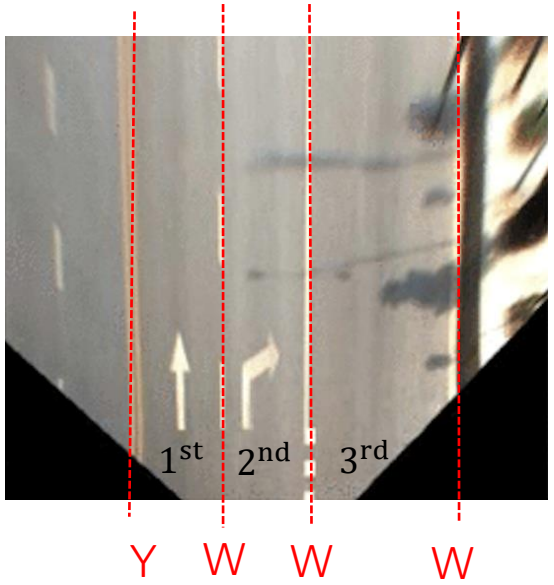


Bird view



Canny

Color of carline information



$$\mathbf{K} = \begin{pmatrix} f\times ku & s & u_0 & 0 \\ 0 & f\times kv & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

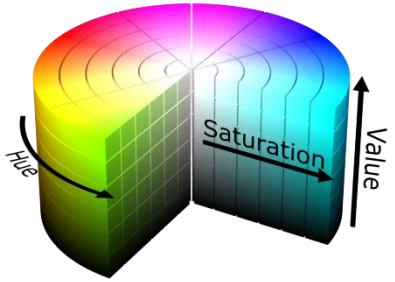
$$\mathbf{T} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -\frac{h}{\sin\theta} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{R} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\langle u, v, 1\rangle^T = \mathbf{KTR}(x, y, z, 1)^T$$

$$\begin{pmatrix} u_i \\ v_i \\ 1 \end{pmatrix} = \begin{pmatrix} p_{11} & p_{13} & p_{14} \\ p_{21} & p_{23} & p_{24} \\ p_{31} & p_{33} & p_{34} \end{pmatrix} \begin{pmatrix} x_w \\ y_w \\ 1 \end{pmatrix}$$

ASSUMPTION $Z_w=0$



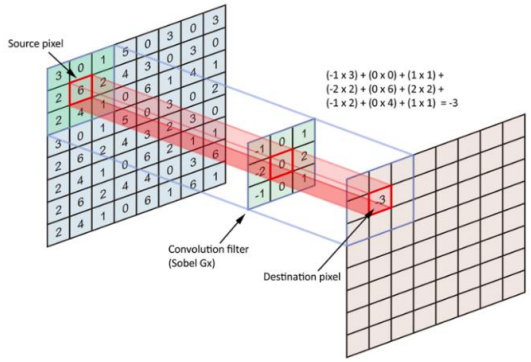
03. RESEARCH PLAN

Architecture / **Perception** / Decision / Control

- *Traffic Sign Detection*

자료: 경찰청

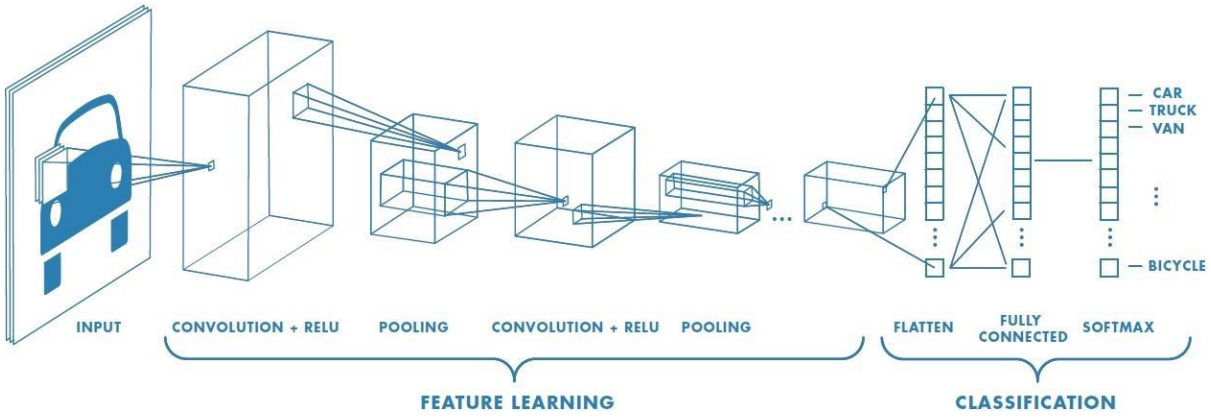
Convolutional neural network



TensorFlow 2.0

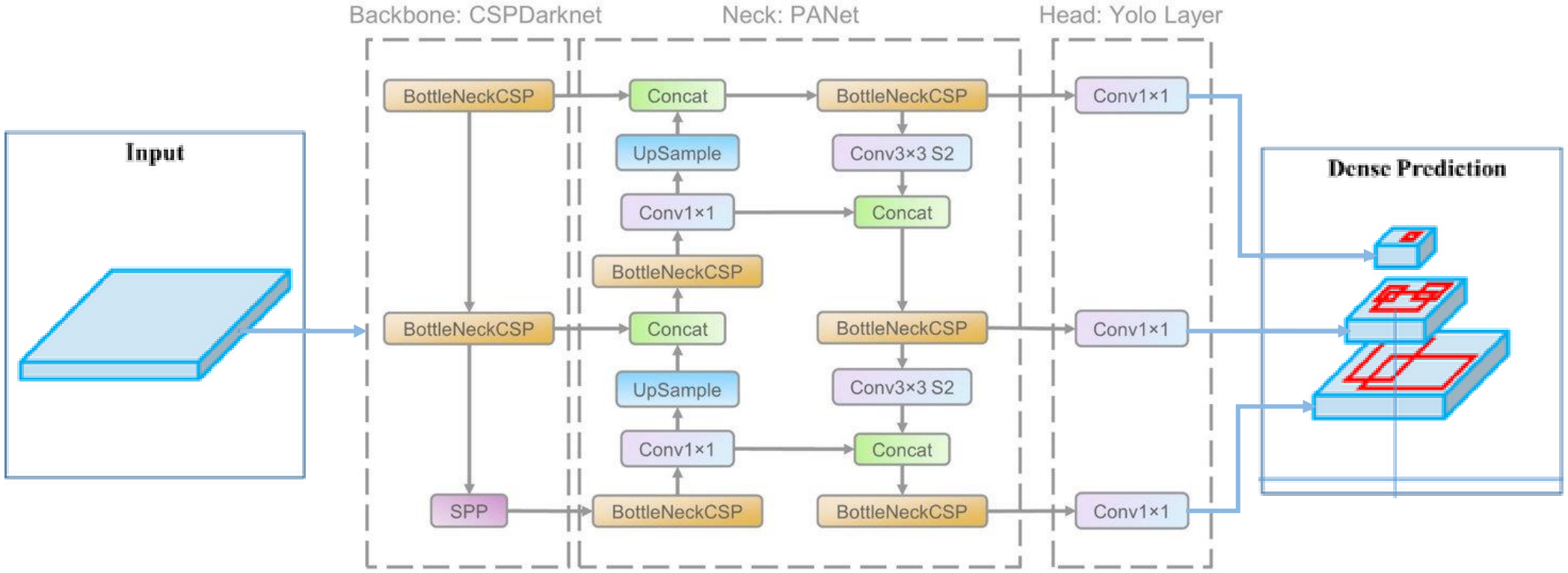
Keras

PyTorch



03. RESEARCH PLAN

Architecture / **Perception** / Decision / Control



Backbone

CSPDarknet 사용, Anchor box 보정
Input image -> Feature map
CLASS 확률 계산

→

Neck

PNet 사용
Backbone 에서 head 사이에서
Feature map 정제, 재구성

→

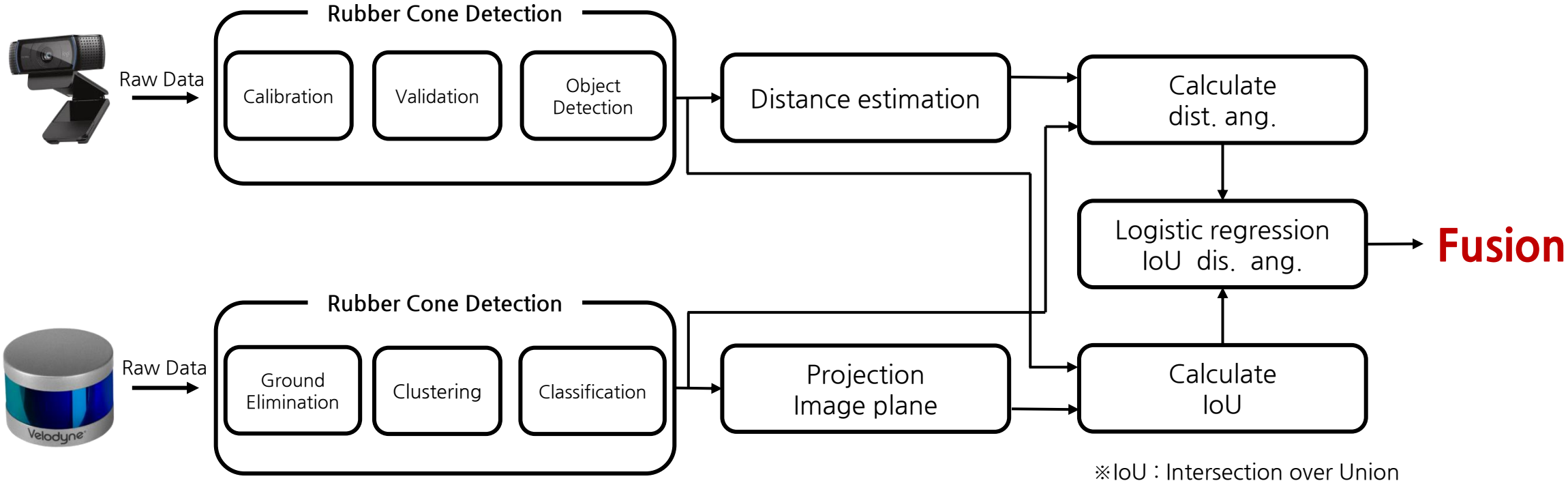
Head

Yolo Layer 사용
PREDICT CLASSES, BOUNDING BOX
REGRESSION

03. RESEARCH PLAN

Architecture / **Perception** / Decision / Control

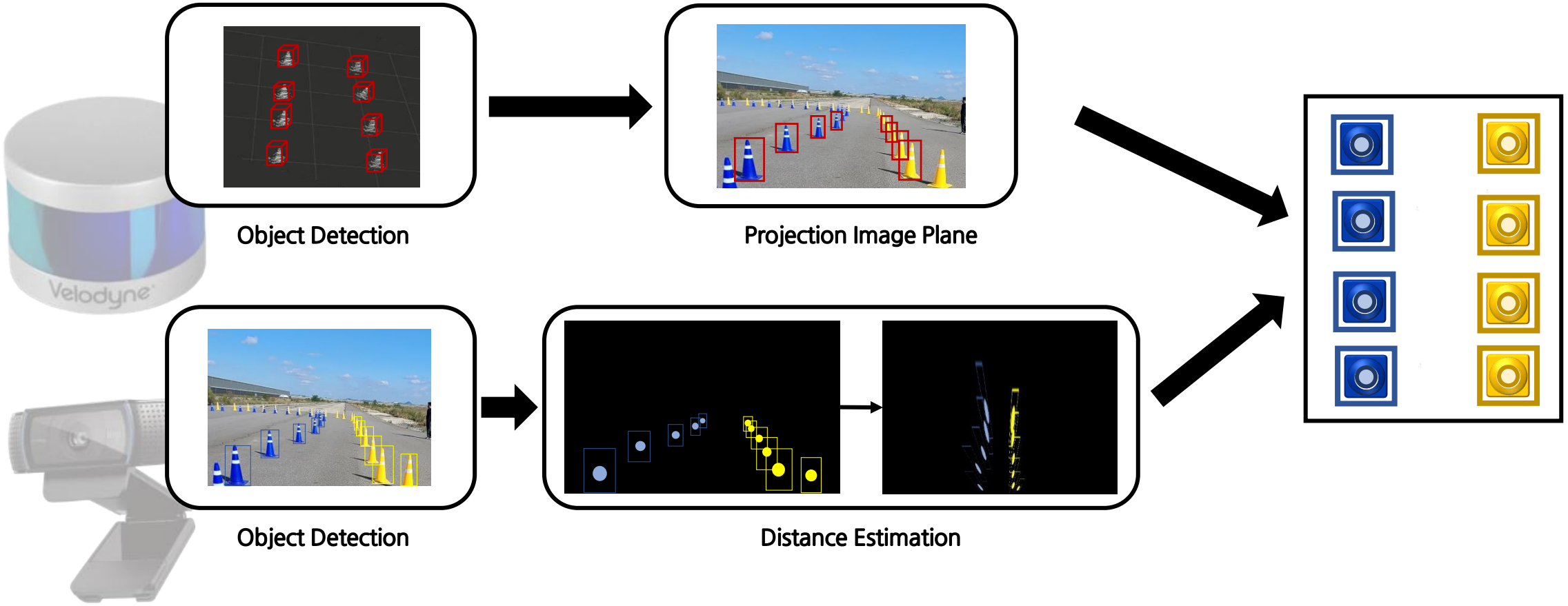
- LiDAR & Vision Fusion



※IoU : Intersection over Union

03. RESEARCH PLAN

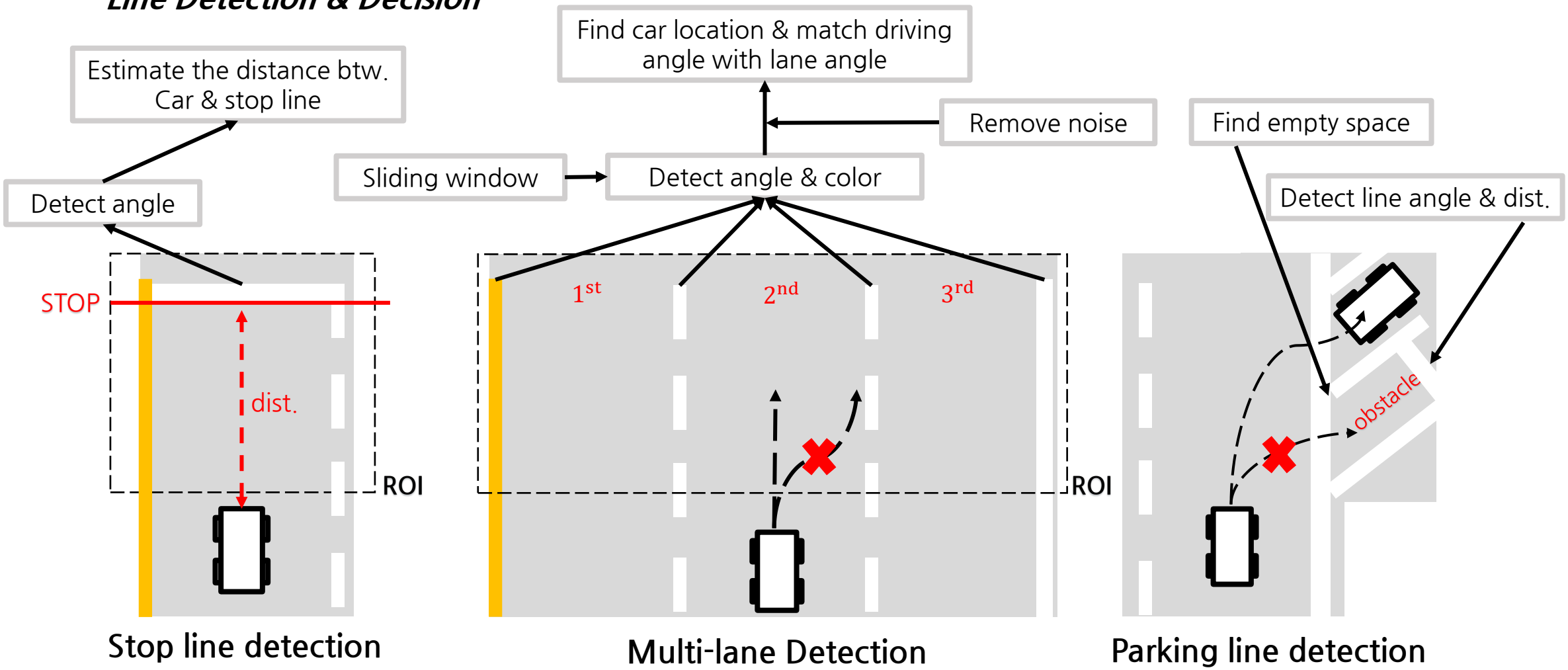
Architecture / **Perception** / Decision / Control



03. RESEARCH PLAN

Architecture / Perception / **Decision** / Control

- *Line Detection & Decision*



03. RESEARCH PLAN

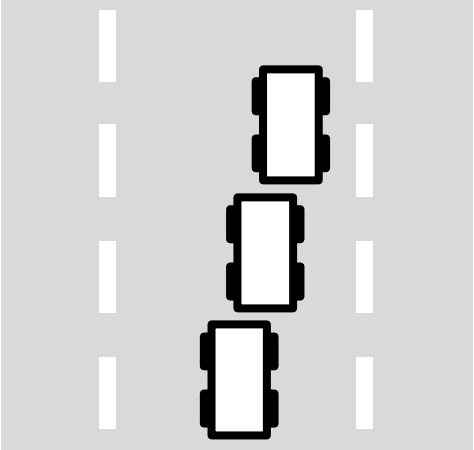
Architecture / Perception / **Decision** / Control

- *ADAS(Advanced Driver Assistance System)*

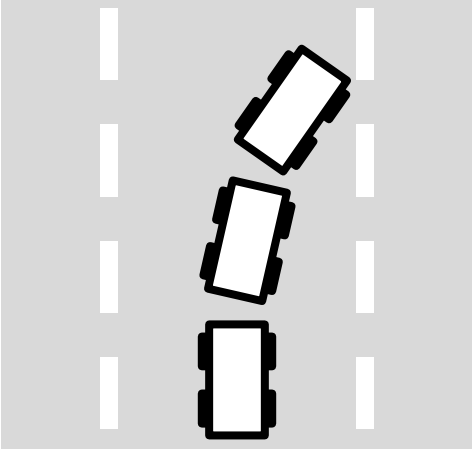
LKAS(Lane Keeping Assist System)

차선 유지 보조 시스템
차량이 주행 중인 차선을 벗어났을 때 기존 차선으로 복귀하는 제어 장치

Case 1. 차선 이탈 발생 가능성



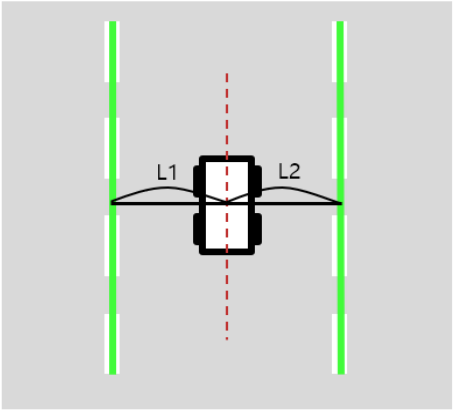
Case 2. 갑작스런 조향 변화



“ 머신비전 카메라를 이용한 실시간 차선 인식을 통해 현재 주행상태 판단 ”

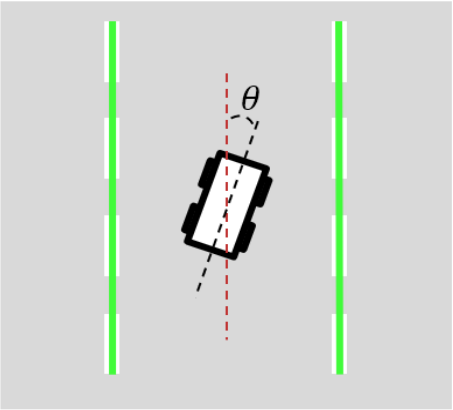
03. RESEARCH PLAN

Architecture / Perception / **Decision** / Control



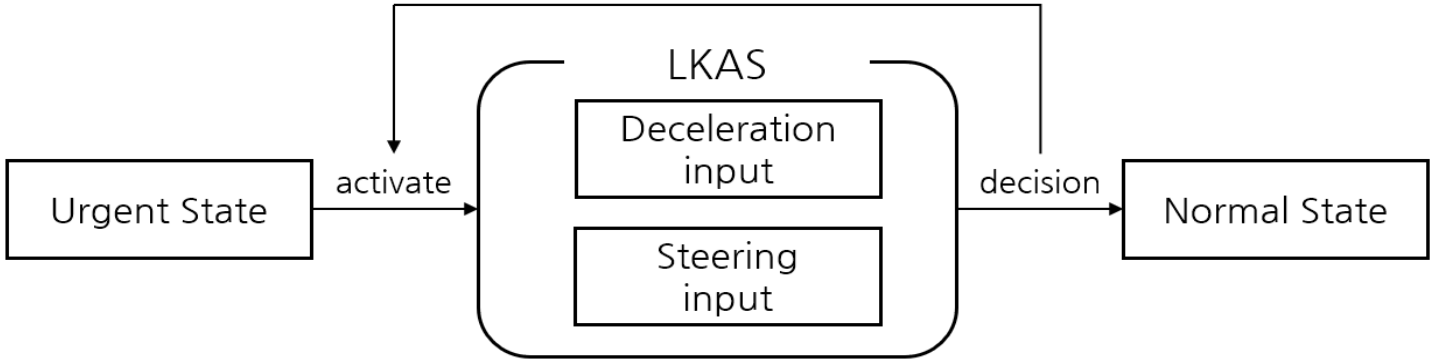
실시간으로 차량과 좌우 차선과의 거리 계산

$$|L1 - L2| \begin{cases} < 50cm & \text{Normal State} \\ \geq 50cm & \text{Urgent State} \end{cases}$$



실시간으로 차량과 좌우 차선과의 각도 계산

$$\theta \begin{cases} < 10° & \text{Normal State} \\ \geq 10° & \text{Urgent State} \end{cases}$$

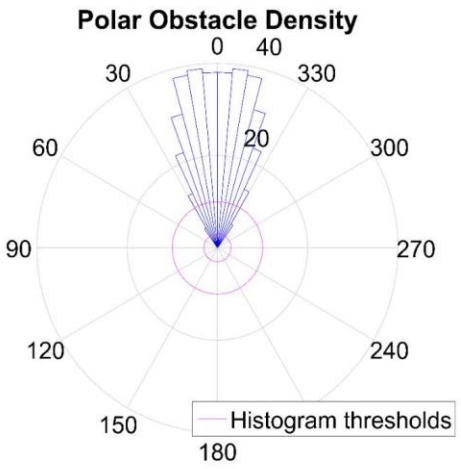


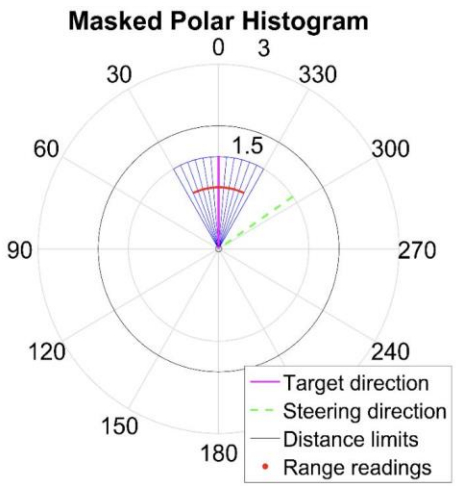
03. RESEARCH PLAN

Architecture / Perception / **Decision** / Control

- Local Path : Obstacle Avoidance Path

→ VFH*





VFH (Vector Field Histogram)

+

4	3	4	5	6		14	15	16	17
3	2	3	4	5		13	14	15	16
2	1	2	3	4		12	13	14	15
1	0	1	2	3		11	12	13	14
2	1	2	3	4		10	11	12	13
3	2	3	4	5		9	10	11	12
4	3	4	5	6	7	8	9	10	11
5	4	5	6	7	8	9	10	11	12

A* 알고리즘

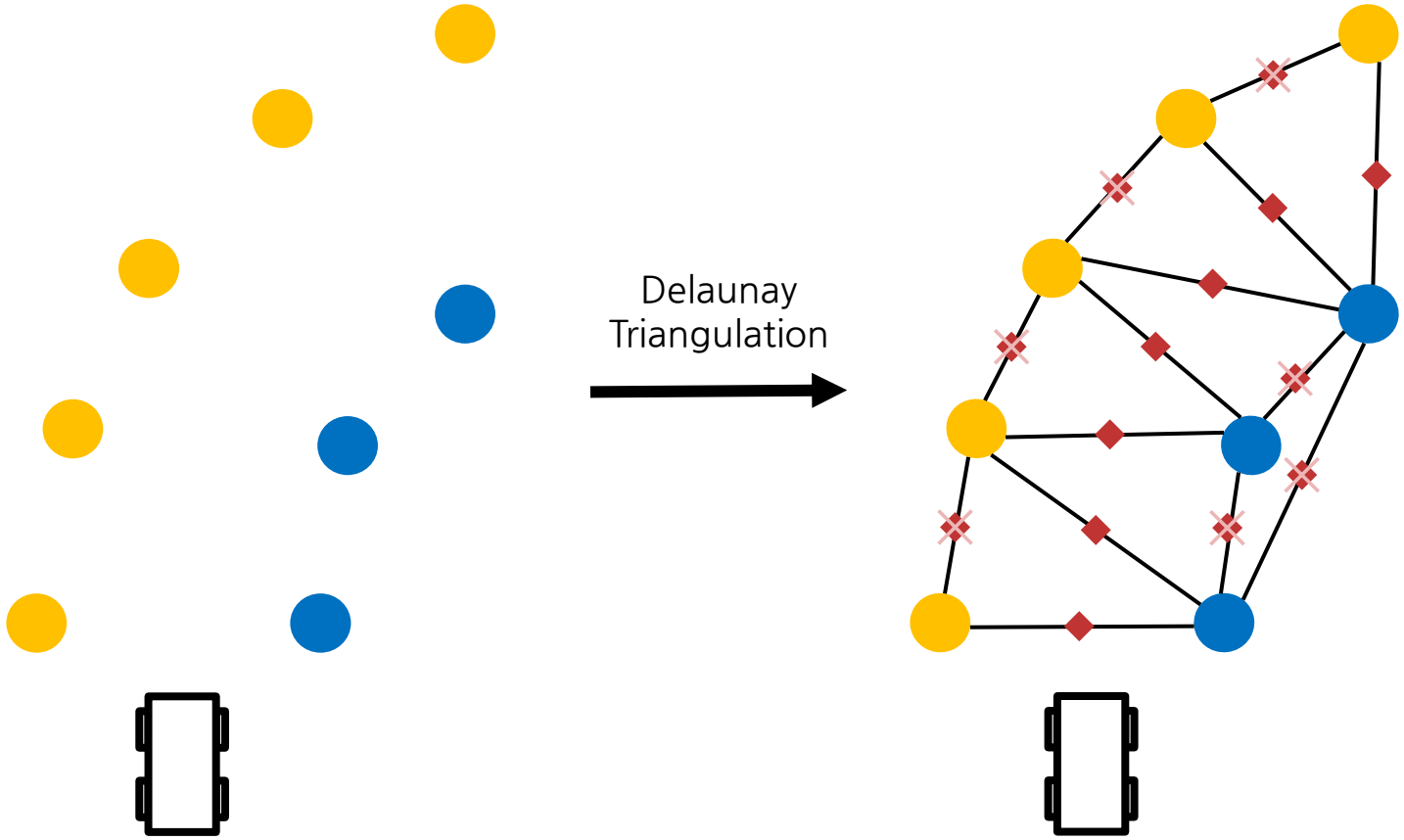
➔ 조향 방향

03. RESEARCH PLAN

Architecture / Perception / **Decision** / Control

- Track : Path Planning

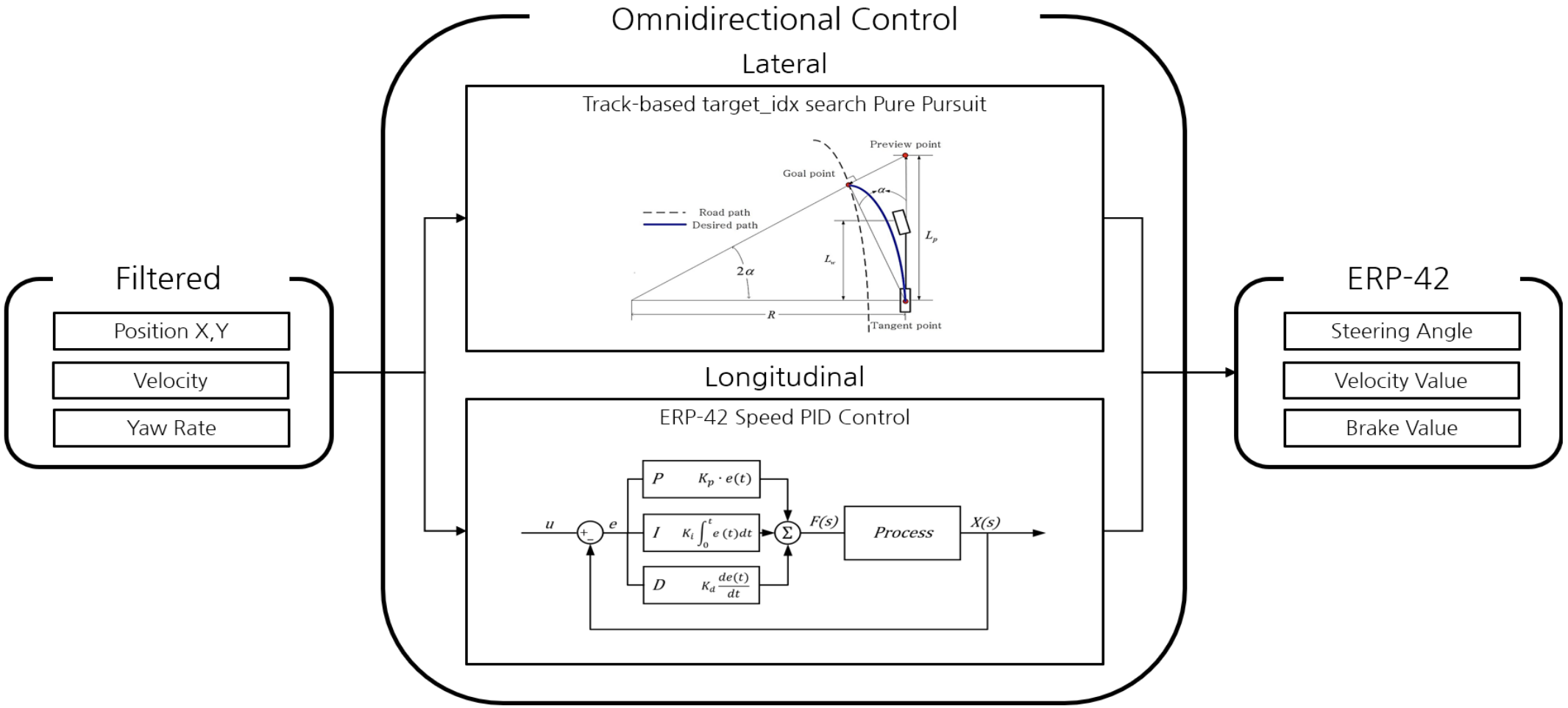
Rubber Cone Position & Color Data



03. RESEARCH PLAN

Architecture / Perception / Decision / **Control**

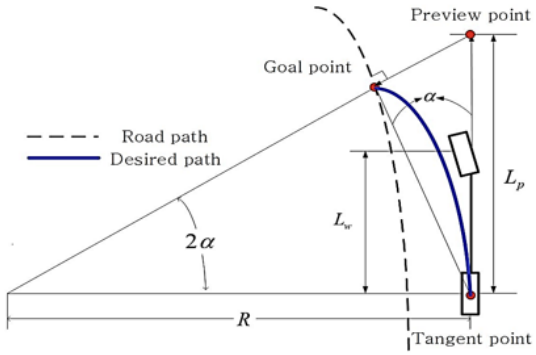
- *Navigation Driving Control*



03. RESEARCH PLAN

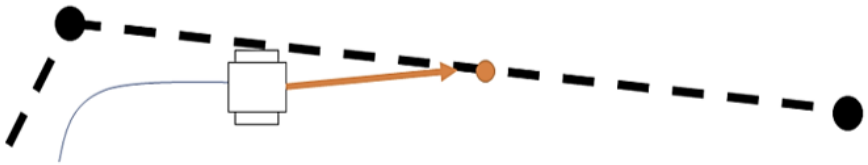
Architecture / Perception / Decision / **Control**

- Lateral : Track-based target_idx search Pure Pursuit



Large Look-Ahead Distance

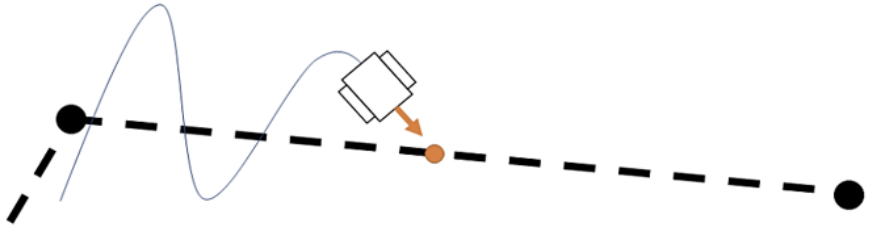
조향각 동작이 부드럽고 차량의 거동이 안정적



과도하게 긴 경우, 곡선 경로에서 Cut-corner

Small Look-Ahead Distance

신속한 경로 복귀가 가능하고 높은 추종 성능

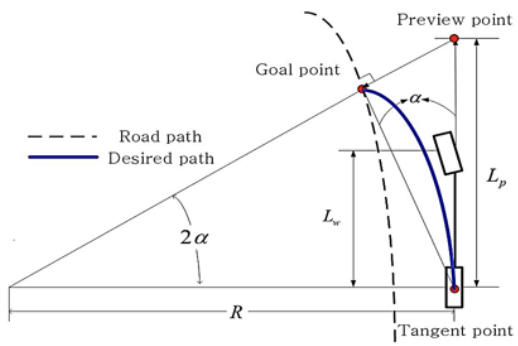


과도하게 짧을 경우, 직선 경로에서 Over-shoot

03. RESEARCH PLAN

Architecture / Perception / Decision / **Control**

- Lateral : Track-based target_idx search Pure Pursuit



$$\frac{ld}{sin2\alpha} = \frac{R}{\sin(\frac{\pi}{2} - \alpha)}$$

$$\frac{ld}{sin\alpha} = 2R$$

$$\frac{ld}{2sin\alpha cos\alpha} = \frac{R}{\cos(\alpha)}$$

$$k = \frac{1}{R} = \frac{2sin\alpha}{ld}$$

$$R = L/tan\delta$$

$$\delta = artan\left(\frac{2Lsin\alpha}{ld}\right) = artan(Lk)$$

➡ 현위치와 목표점 간의 곡률이 조향각 산출의 핵심 파라미터

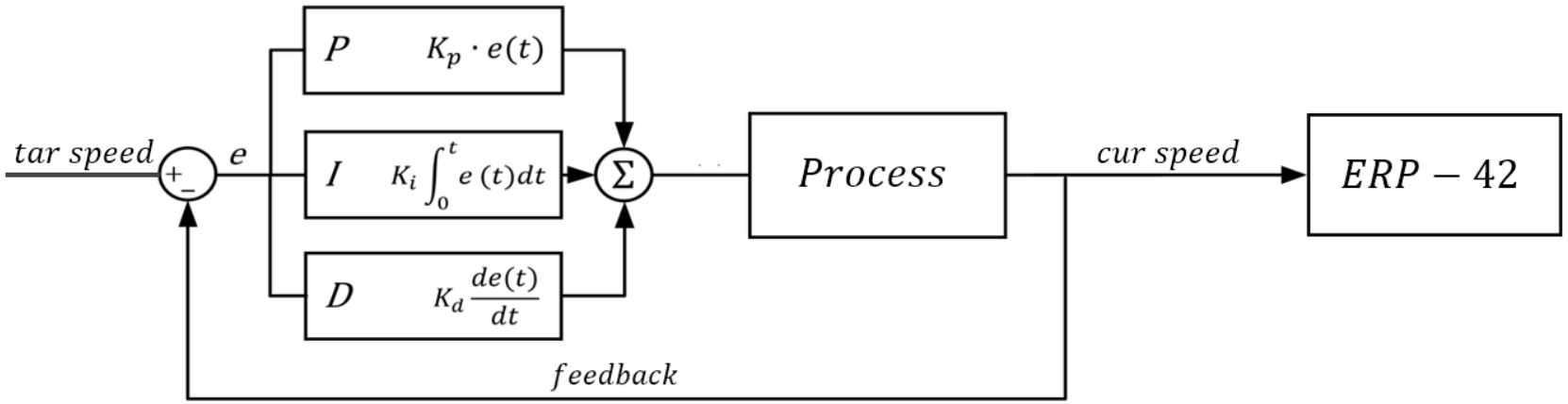
주행 경로 구간 별 곡률에 따른 예견거리 설정 및 목표점 탐색 Pure Pursuit

RDDF → X,Y → 구간별 곡률 계산 → Ld → 곡률 기반 목표점 선정 → Tar_idx → Pure Pursuit → δ → ERP-42

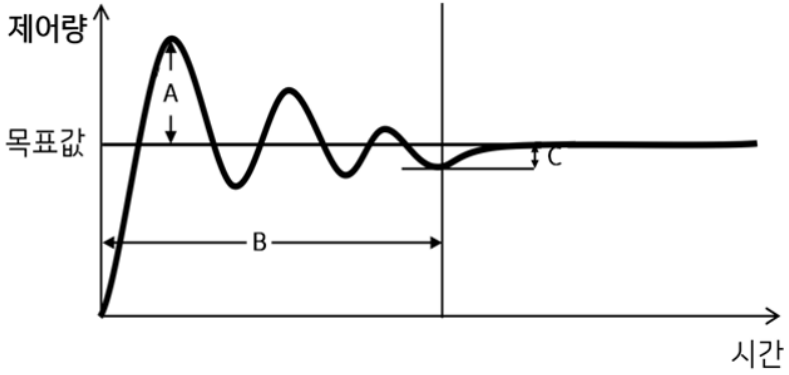
03. RESEARCH PLAN

Architecture / Perception / Decision / **Control**

- Longitudinal : ERP-42 Velocity PID Control



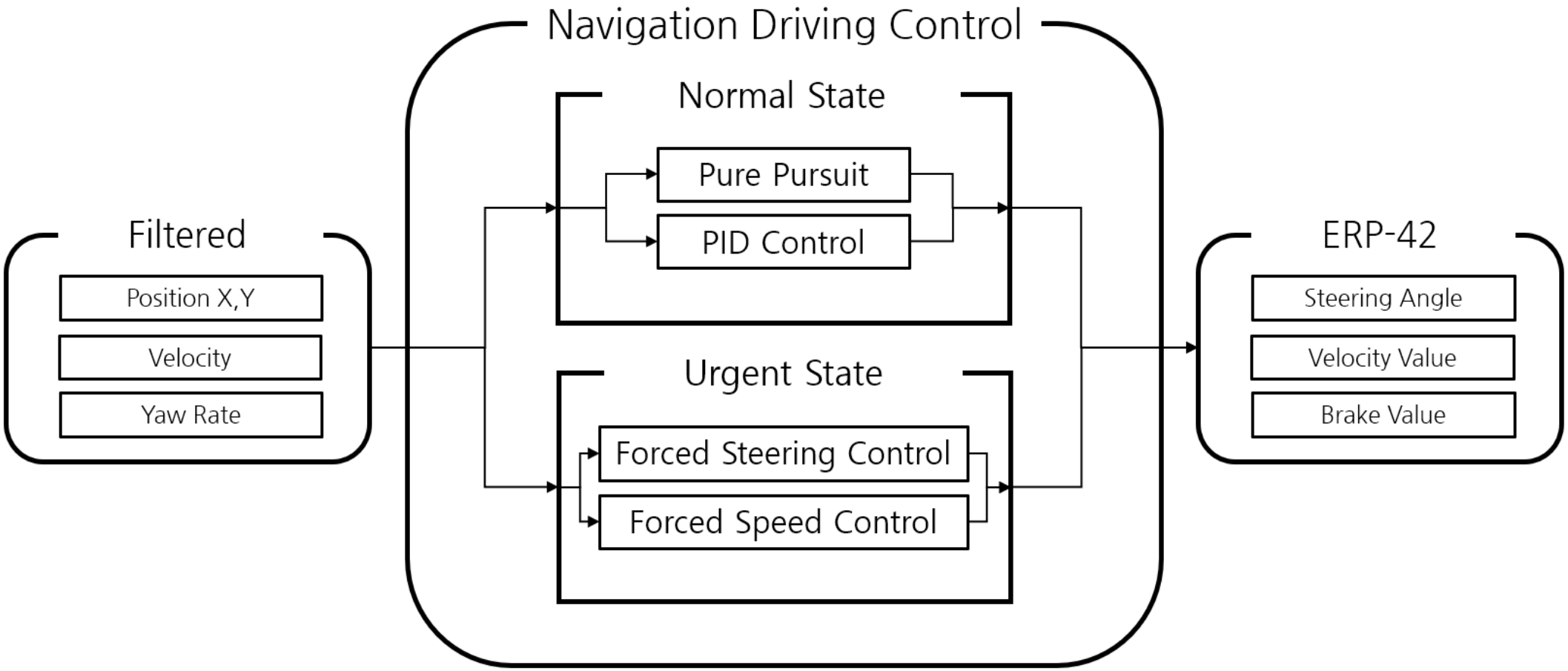
- P제어 : 목표값 도달 시간(B) 감소 - 목표값과 멀수록 많이, 가까울수록 적게 제어
- I제어 : 정상 상태 오차(C) 감소 - 목표값에 근접할 수 있도록 미세하게 제어
- D제어 : 과도 Over-shoot(A) 억제 - 제어값 변화 억제



03. RESEARCH PLAN

Architecture / Perception / Decision / **Control**

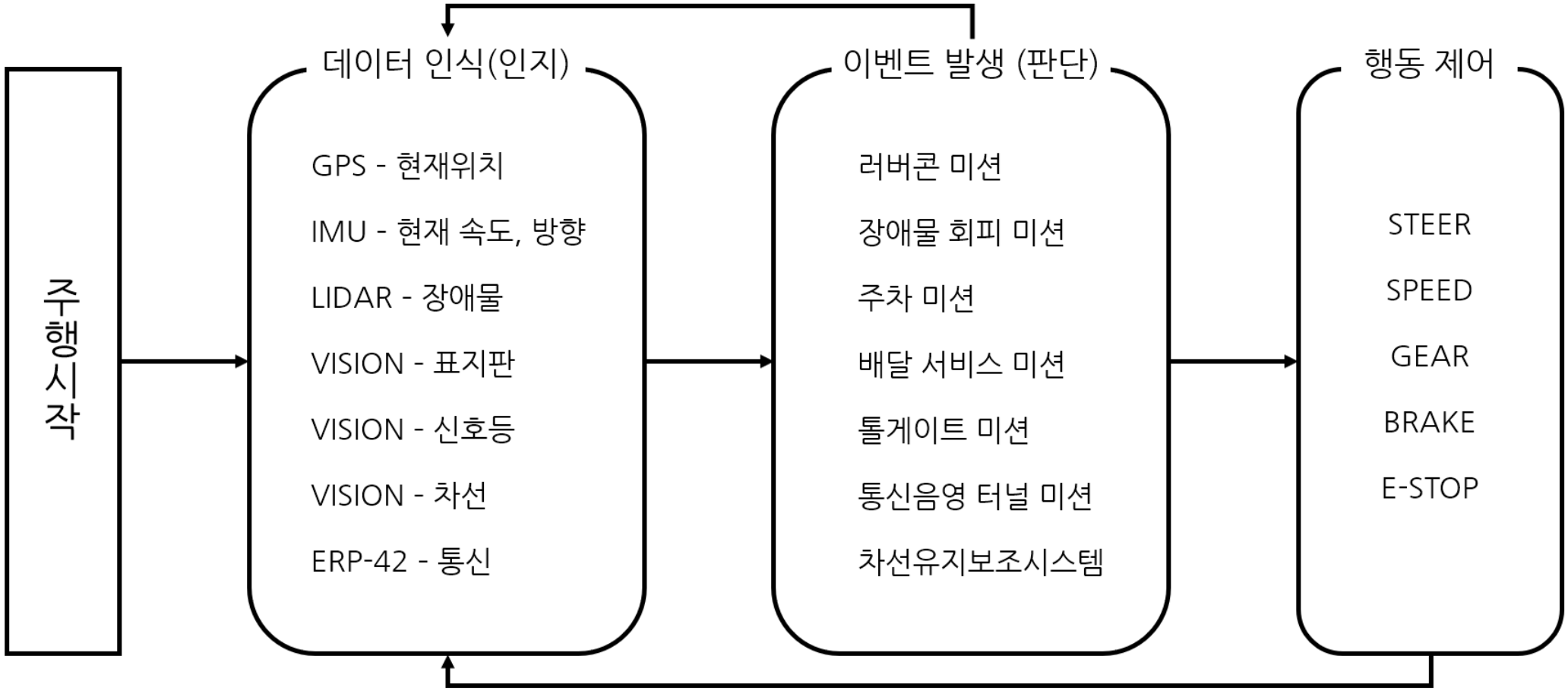
- *Navigation Driving Control*



03. RESEARCH PLAN

Architecture / Perception / Decision / **Control**

- State Machine



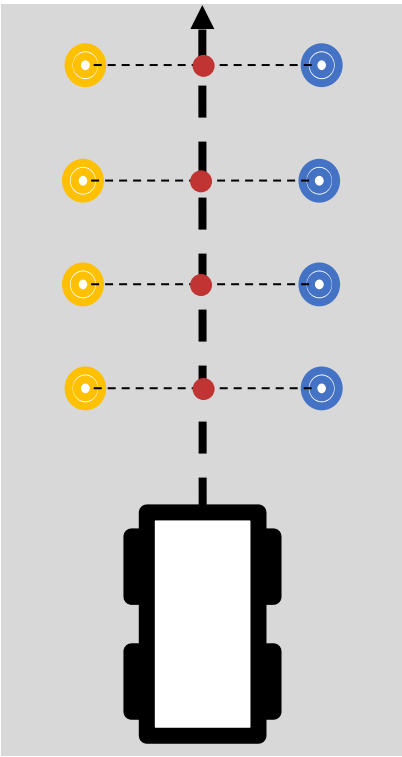
04. MISSION ALGORITHM

Track Driving / Obstacle Avoidance / Parking / Delivery / Highway

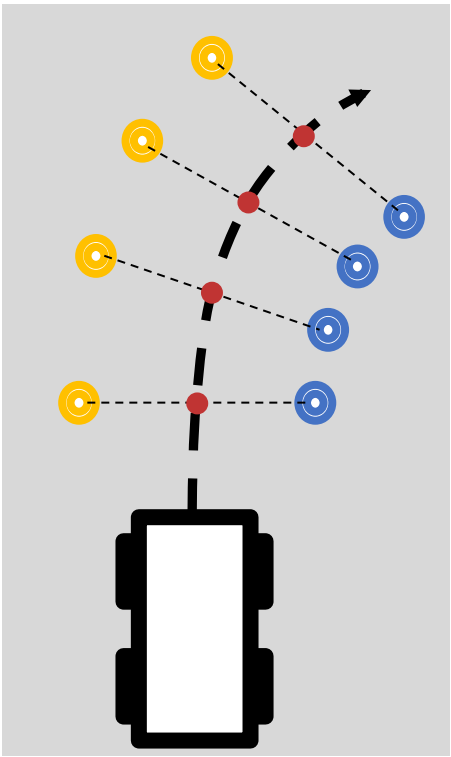
04. MISSION ALGORITHM

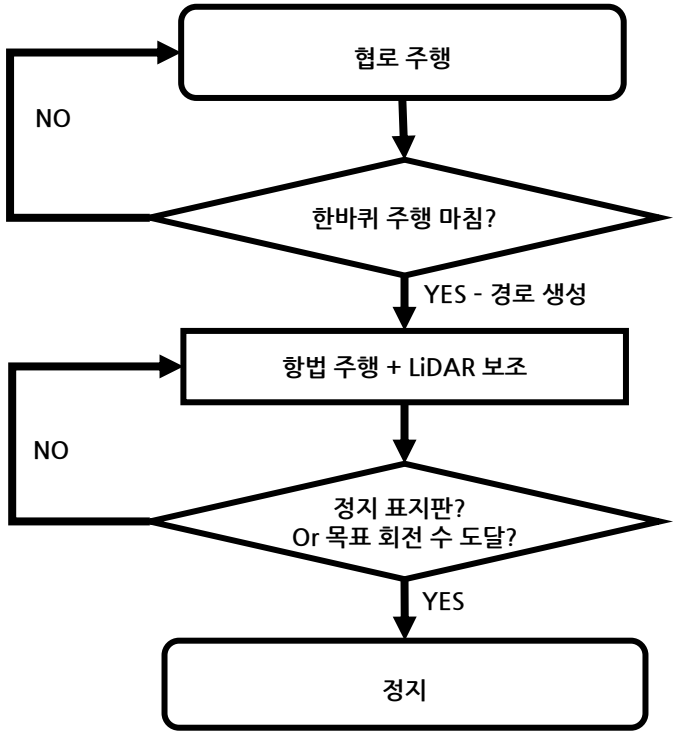
Track / Obstacle Avoidance / Parking / Delivery / Highway

- 직진 코스



- S 커브 코스

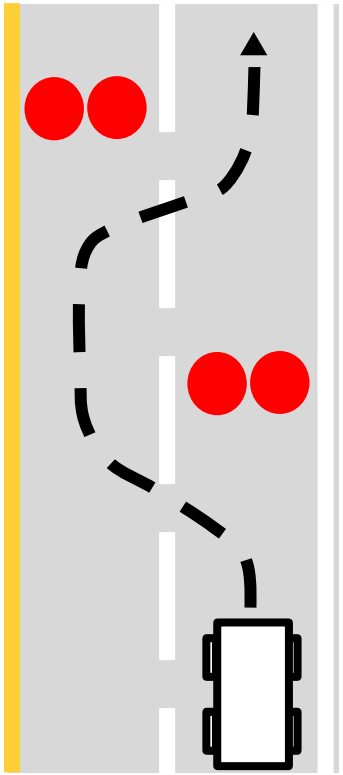


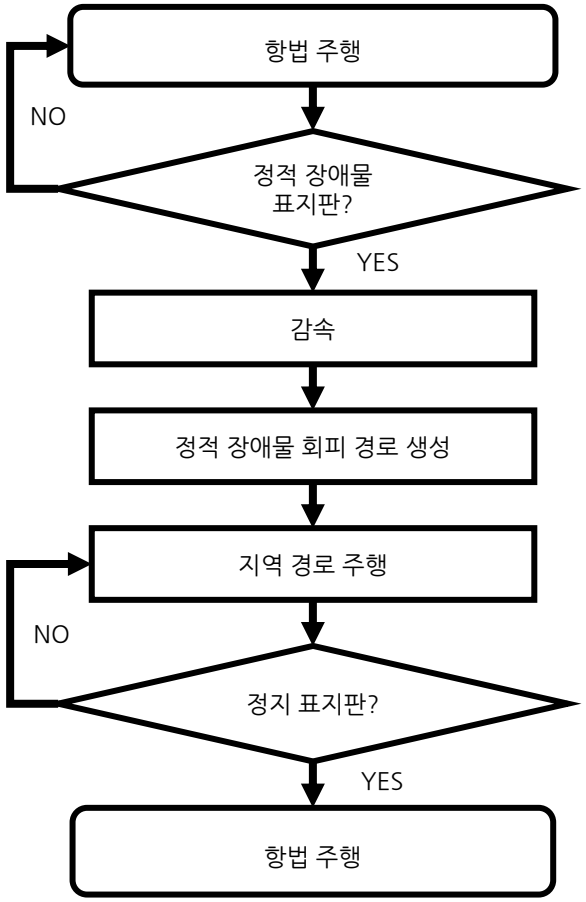


04. MISSION ALGORITHM

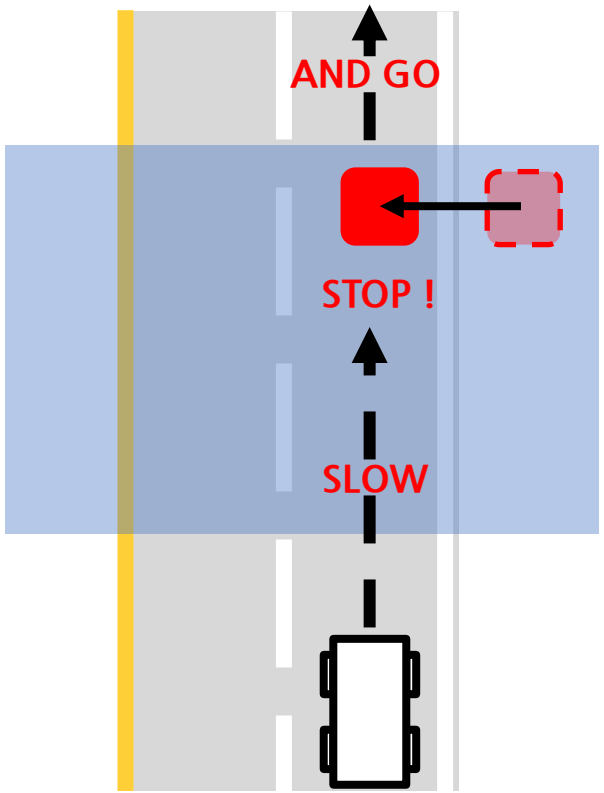
Track / **Obstacle Avoidance** / Parking / Delivery / Highway

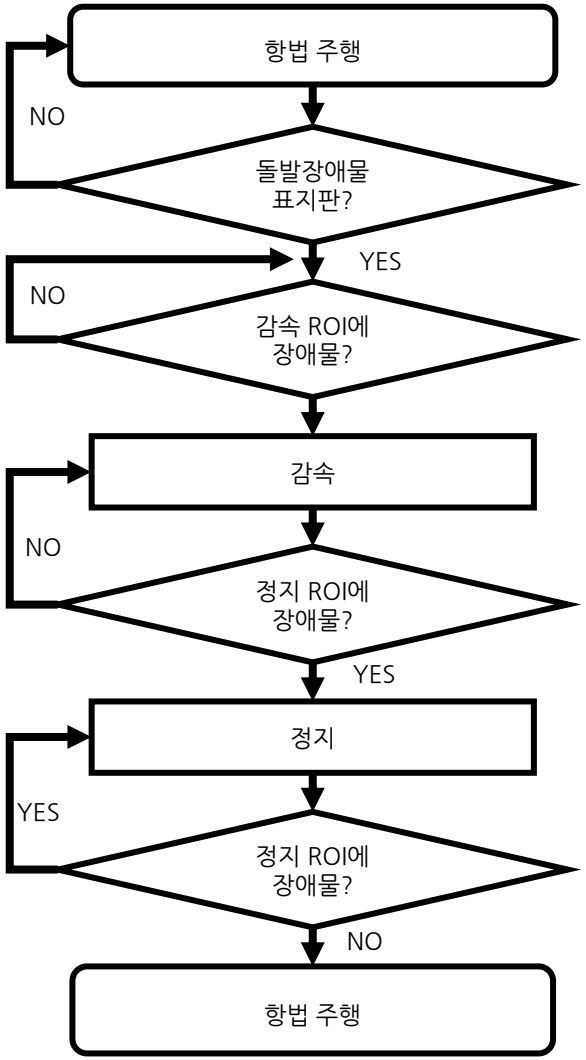
- 정적 장애물 회피





- 동적 장애물 회피

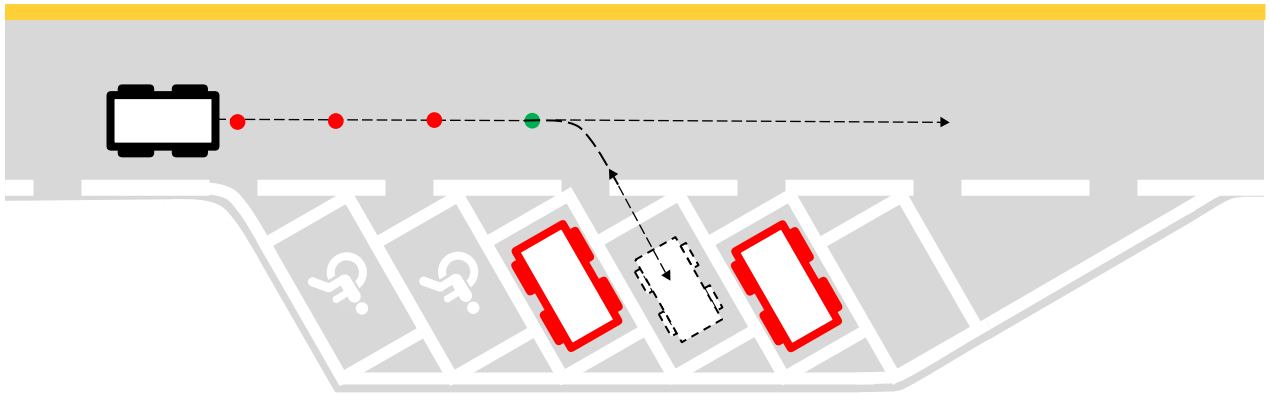




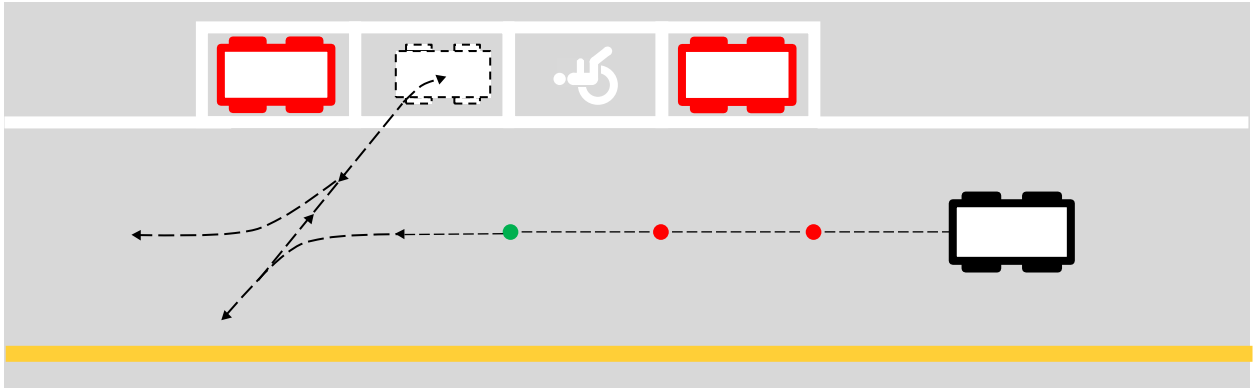
04. MISSION ALGORITHM

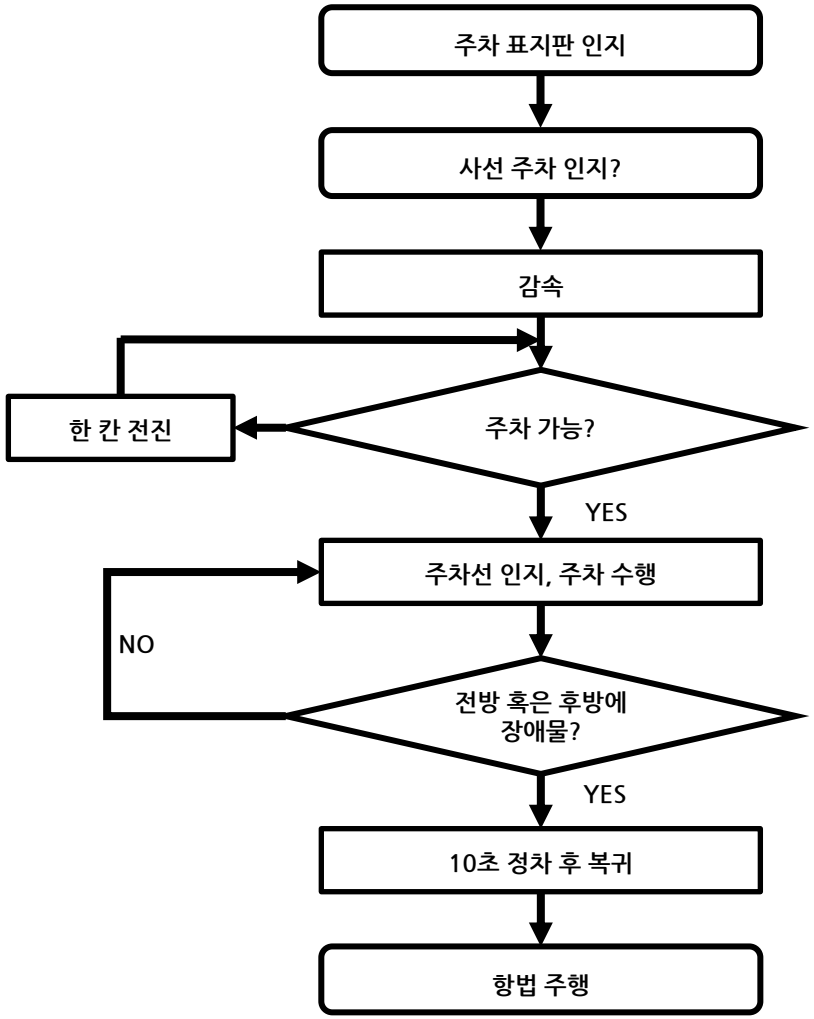
Track / Obstacle Avoidance / **Parking** / Delivery / Highway

- 사선 주차



- 평행 주차

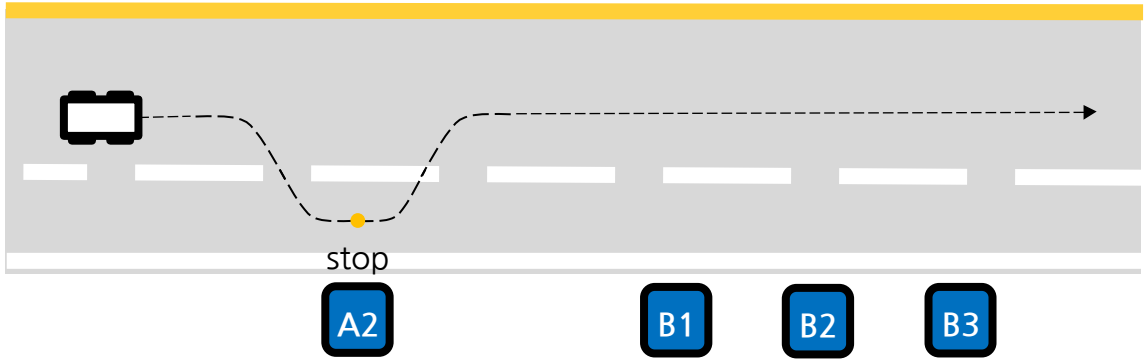




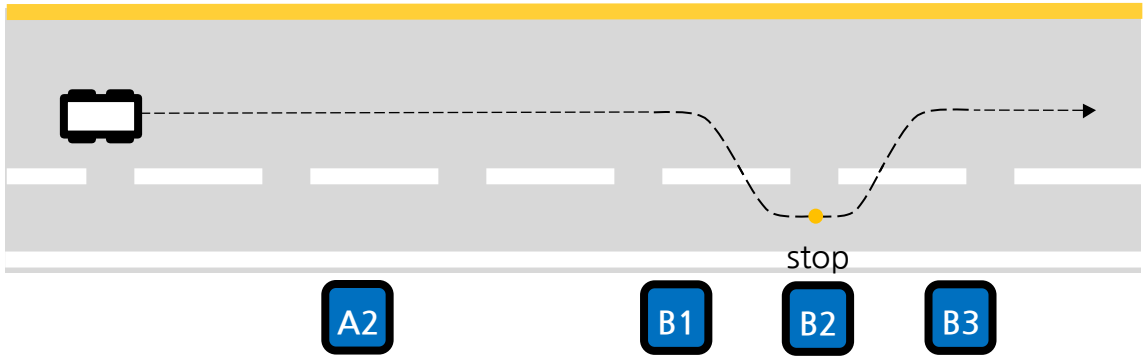
04. MISSION ALGORITHM

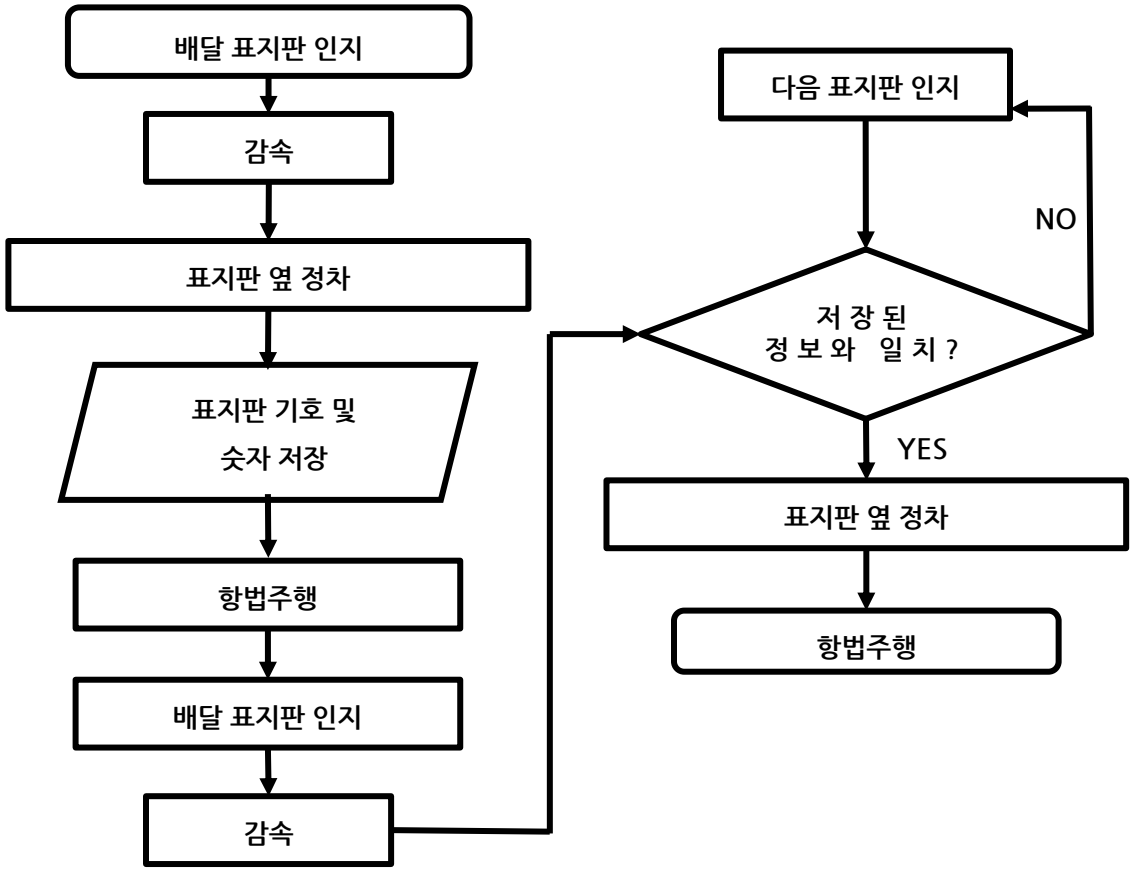
Track / Obstacle Avoidance / Parking / **Delivery** / Highway

- 배달 픽업



- 배달 전달

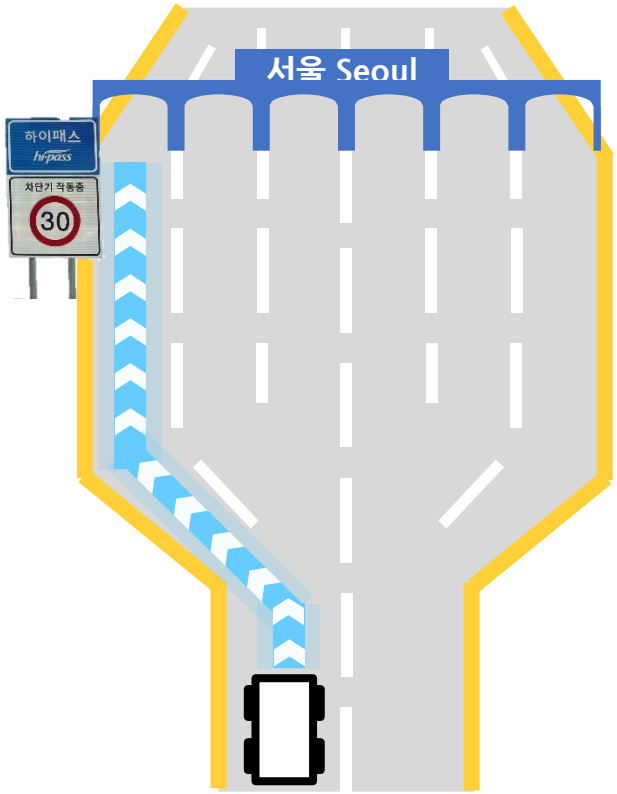




04. MISSION ALGORITHM

Track / Obstacle Avoidance / Parking / Delivery / **Highway**

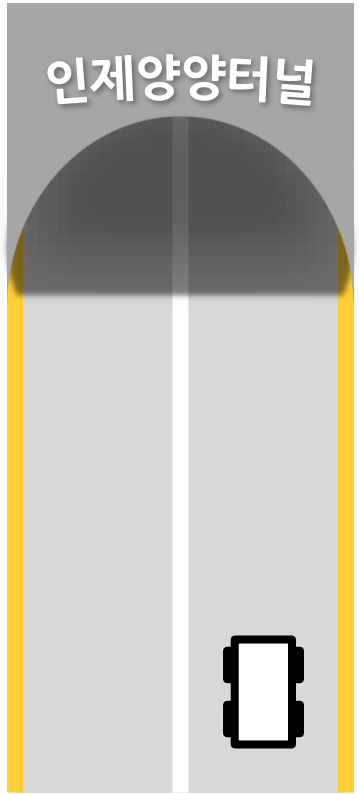
- 톨게이트



- 카메라를 **이용해 바닥 유도선** 및 하이패스 표지판 인식

- 하이패스 인식이 가능하도록 톨게이트 접근 시 **감속**

- 터널 (음영구역)



- IMU를 사용하여 터널 내의 현재 위치 추정

- 터널 진입 시 변하는 **조명 상태 (햇빛 유무, 터널 조명 등) 에 따른 카메라 임계값** 조정

05. ADDITIONAL INFO.

Budget / Development Schedule

05. ADDITIONAL INFO.

Budget / Development Schedule

제품명	단가	수량	금액
[Xsens] MTi-7-0I-DK (IMU)	860,000	1	860,000
[YDLiDAR] X4 Pro (2D LiDAR)	93,500	1	93,500
Machine Vision CA model (MVC)	500,000	1	500,000
머신비전 카메라 렌즈	87,000	1	87,000
볼 마운트 브라켓	16,800	4	67,200
러버콘 (노랑)	6,000	30	180,000
러버콘 (파랑)	5,080	30	152,400
사전연습주행 용달비 (시흥-화성 왕복)	200,000	8	1,600,000
플랫폼(ERP-42) 수리비	견적 문의 예정	-	약 2,500,000
알루미늄 프로파일	주문 제작 예정	-	약 196,000
외장 재료비용	주문 제작 예정	-	약 300,000
합계			6,536,100

05. ADDITIONAL INFO.

Budget / **Development Schedule**

HONGIK UNIV.

Team	항목	연구일정 (월)					
		5	6	7	8	9	10
Control & GPS	Filtered GNSS DATA PARSING						
	곡률정보 기반 Path Planning						
	종방향 PID 속도 제어						
	State Machine 구축						
LiDAR	정적 및 돌발 미션						
	주차 미션						
LiDAR & Vision	트랙 주행						
Vision	신호등, 표지판 및 라바콘 학습						
	차선 유지 보조 알고리즘						
	주행 및 미션 관련 파라미터 최적화						
Hardware	카울 및 센서 부착						
Integration	코드 통합 및 미션 수행 실험						

THANK YOU